

Charles Morin

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Education

McGill University

- Master of Science – Mechanical Engineering 2026-Present
- Bachelor of Mechanical Engineering, GPA: 3.92/4.00 2022 – 2026

Awards

McGill University

- Dean's Honour List 2026
- Summer Undergraduate Research in Engineering Award - NSERC USRA 2026
- EUS Club of the Year - McGill Calisthenics Club 2026

Skills

Software: CAD: CREO, SolidWorks, NX, FEA: Abaqus, CFD: ANSYS CAM: Mastercam, Life Cycle Library: PLM
Programming: React (100h course), Python, C#, JavaScript, Node.js, HTML, CSS, Flutter, Dart, MongoDB, R
Others: MS Office, MatLab, Arduino
Languages: French (Fluent), English (Fluent), Spanish (beginner)

Experience

CAE, Mechanical Design – Hardware Engineering

May – August 2025

Immersive Ship Simulator Upgrade

- Led the design of a new visual installation in close collaboration with the client, ensuring solutions not only met but exceeded expectations. Work included a surrounding display installation and custom bracket assemblies
- Collaborated with designers, leads, and installation crews on lifting equipment installations which required extension platforms and custom tools to lift projectors
- Developed electrical equipment installation in collaboration with team members, including custom cabinets, covers, and bracketry for a new projector system
- Designed a wide range of additional assemblies and components - from adjustable blend plate holders to safety equipment installation (custom locks and safety-zone demarcation)
- Presented designs regularly to design leads, technical leads, and floor workers, welcoming feedback crucial to assure the success of the project

CAE, Mechanical Design – Hardware Engineering

May – December 2024

Mechanical Safety Supports for Ladder

- Led the design of a ladder and its supports for an inclined simulator while meeting deadlines
- Planned and researched multiple ideas for optimal solutions meeting design criteria such as low cost, ease of manufacturing, simplicity to use and easy installation
- Designed 10 supports and analyzed stresses of ladder using Creo from start to finish
- Presented, design reviewed, optimized and documented in collaboration with other designers and reviewers using PLM (Product Lifecycle Management)

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McGill Capstone Project – ARA Robotics

Fall 2025 – Winter 2026

- Designed a folding joint mechanism enabling the drone arms to collapse for transport
- Analyzed stress concentrations and load distribution in the assembly using ANSYS under in-flight, landing, and transportation load cases
- Optimized the joint to meet client requirements, including weight, aesthetics, manufacturability, assembly simplicity, structural strength, and usability
- Manufactured and assembled working prototype

McGill Formula Electric (MFE) - Controller Design

Fall 2025

- Designed and tuned a robust controller using D–K iteration with model uncertainty represented in a generalized plant
- Performed system identification of the cooling water pump using least-squares methods and experimental data collected by the MFE team
- Achieved a 69.7% reduction in energy consumption compared to the initial vehicle cooling system

McGill Cardiovascular Mechanics and Biofluids – CFD Simulations of Stenosis in Carotid Arteries

Winter 2026

- Modeled carotid arteries with varying stenosis severities using computed tomography (CT) and volumetric magnetic resonance angiography (MRA) data
- Simulated blood flow through arterial models using CFD in ANSYS Fluent

McGill Computational Gas Dynamics – Development of Numerical Solvers for Euler Equations

Winter 2026

- Developed numerical solvers for the Euler equations using various discretization schemes

McGill Design and Stress Analysis Project

Fall 2024

- Designed a bracket to specific stiffness before manufacturing it using a laser cutter for a competition
- Analyzed stress concentrations, boundary condition, and load applications using Abaqus
- Predicted failure location and load

McGill Design and Manufacturing Project

Winter 2024

- Designed and manufactured a nozzle, mold, and side action using different manufacturing techniques such as 3D printing in combination with SolidWorks, milling and turning with Mastercam

McGill Robotics Project

Winter 2024

- Achieved 2nd place in competition and 1st place in time
- Led team to build a lightweight robot that completed the obstacle course in minimal time

Extracurricular

Vice President Internal — McGill's Calisthenics Club

2025-2026

Member — McGill's Dragon Boat Team

2025-2026

Member — McGill Poker Club

2026

Volleyball player — McGill Intramural League

2026

Basketball player — McGill Intramural League

2025

Weightlifting, Tennis, Ski, Snowboard, Golf, Reading

Lifelong